ACME AVIATION MAINTENANCE

Hangar Operations Manual

Revision 3 | 2024-01-15

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1. HANGAR SAFETY AND ACCESS CONTROL

All personnel entering maintenance hangars must check in at the security desk and display a valid ACME Aviation badge at all times. Visitors must be escorted by authorized maintenance staff.

Personal protective equipment (PPE) requirements:
- Safety glasses (ANSI Z87.1 rated)
- Steel-toe or composite-toe boots
- High-visibility vest when on the ramp
- Hearing protection in designated noise areas

No loose clothing, jewelry, or lanyards near rotating equipment. Long hair must be secured. Cell phone use is prohibited in designated fuel handling and painting areas.

Speed limit inside hangars: 5 MPH for all vehicles and tugs. Pedestrians have the right of way at all times.

Emergency exits are marked with illuminated signs. Familiarize yourself with the evacuation route posted at each entrance. Assembly points are located in Parking Lot C and Lot D.

2. TOOL CONTROL PROGRAM

All tools used on aircraft must be inventoried before and after each maintenance task. Use shadow boards and foam inserts in tool boxes to easily identify missing items.

If a tool cannot be accounted for at task completion:

1. Stop work immediately
2. Notify the Lead Technician
3. Conduct a thorough search of the work area
4. Inspect the aircraft using borescope if necessary
5. Complete Lost Tool Report (Form TC-50)

The aircraft SHALL NOT be released for flight until all tools are accounted for or the aircraft has been inspected and cleared by the Quality Inspector.

Personal tools are not authorized for use on aircraft unless individually approved and etched with the technician's ID.

Calibrated tools must display a current calibration sticker. Check calibration dates before use. Report expired tools to the Tool Room attendant.

WARNING

Handling Fuel

Authorized Personnel Only

3. FUEL HANDLING PROCEDURES

All fuel operations must comply with NFPA 407 and the airport authority fuel handling regulations. Key requirements:

- Ground all equipment before connecting fuel hoses
- No smoking or open flames within 100 feet of fueling
- Wear anti-static PPE during all fuel operations
- Maximum flow rate: 300 GPM underwing, 50 GPM overwing
- Monitor fuel temperature (suspend if >40C / 104F)
- Two-person rule for all overwing fueling operations

Fuel spills >5 gallons: activate alarm, call AOC ext. 9000, deploy absorbent booms. Do NOT wash into drains.

Report all spills on Form IR-200 within 2 hours.

4. HAZARDOUS MATERIALS MANAGEMENT

All hazardous materials used in maintenance operations must be stored in approved, labeled containers in the HazMat locker. Safety Data Sheets (SDS) for all materials are maintained in the red binder at each work station and digitally on the intranet.

Common hazardous materials in aircraft maintenance:
- Aviation fuels (Jet-A, 100LL Avgas)
- Hydraulic fluid (MIL-PRF-83282)
- MEK (Methyl Ethyl Ketone) solvent
- Chromate primers and coatings
- Battery electrolyte (sulfuric acid / KOH)
- Sealants and adhesives

Dispose of all hazardous waste in designated containers. Never pour chemicals down drains. Waste oil, fuel, and solvents must be collected separately.

Spill kits are located at each hangar entrance (yellow cabs). HazMat team contact: ext. 4500 (24/7).

5. GROUND SUPPORT EQUIPMENT

Ground Support Equipment (GSE) includes all motorized and non-motorized equipment used to service aircraft on the ground.

GSE Inspection Requirements:
- Daily: fluid levels, tire condition, lights, brakes
- Weekly: battery connections, safety switches, fire ext.
- Monthly: full operational test per GSE manual
- Annually: comprehensive inspection and recertification

Only trained and authorized personnel may operate GSE. Complete GSE Operator Training (course GT-200) before use.

Report all GSE defects on Form GSE-100. Tag defective equipment with a red OUT OF SERVICE tag and move to the GSE maintenance area.

Maximum approach speed to aircraft: 2 MPH.
Always use a wing walker when positioning GSE near aircraft.
Maintain 10-foot clearance from aircraft when not in use.

6. AIRCRAFT JACKING PROCEDURES

Aircraft jacking is a critical operation that requires a minimum of three qualified personnel: one jack operator per jack and one safety observer. The Lead Technician must authorize all jacking.

Before jacking:

1. Verify aircraft weight does not exceed jack capacity
2. Position jacks at approved jack points per AMM
3. Install tail stand if required by aircraft type
4. Ensure hangar doors are closed (wind restriction: 15 kt)
5. Disconnect external power and ground equipment
6. Brief all team members on the jacking plan

During jacking:

- Raise all jacks simultaneously at equal rates
- Monitor aircraft attitude continuously with spirit level
- Stop and investigate any unusual sounds or movement
- Install safety collars on all jacks when target height reached

Never walk under a jacked aircraft unless safety jacks are set.

7. CONFINED SPACE ENTRY

Confined spaces on aircraft include fuel tanks, cargo holds, wheel wells, and tail cone areas. Entry requires a Confined Space Entry Permit (Form CS-100) signed by the Safety Manager.

Requirements before entry:
- Atmospheric testing (O2, LEL, CO, H2S)
- Continuous ventilation with fresh air supply
- Standby attendant stationed at the entry point
- Communication system between entrant and attendant
- Rescue plan and equipment readily available

Minimum oxygen level: 19.5%. Maximum LEL: 10%.
If atmosphere becomes hazardous, evacuate immediately.

Fuel tank entry additionally requires:
- Tank fully drained and purged with inert gas
- Electrical bonding of all tools and equipment
- Anti-static clothing and footwear
- 48-hour waiting period after last fuel presence

8. REVISION HISTORY

Rev	Date	Author	Description
3	2024-01-15	J. Rodriguez	Added confined space entry procedures
2	2023-06-01	M. Thompson	Updated fuel handling per NFPA 407
1	2022-11-20	J. Rodriguez	Added tool control program
0	2022-03-01	K. Williams	Initial release

APPROVAL SIGNATURES

Director of Maintenance: _____ Date: _____

Quality Manager: _____ Date: _____

Safety Manager: _____ Date: _____